



DATA-537 NFL Prediction

By: Grant Gonnerman

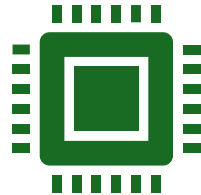
Introduction



The NFL has seen massive growth in recent years.

Expanding into other countries for international games.

Setting new viewership records during holiday games.



AWS along with many other companies are providing real time statistics adding another layer to the game.



I will look at weekly team statistics with the goal of identifying the most important features to predict team wins and losses.

Data Description



I collected this data from Yahoo sports.

There are no ethical concerns with this data because it is all publicly available with no confidential information.



Data was saved into excel files.

Aggregated into team totals with python.



Collected from weeks 1-13 of the 2025 NFL season.



Contains defense, kicking, kickoffs, passing, punting, receiving, returns, rushing, and win/loss data.



416 rows and 79 columns in this dataset and 191 missing values.

Column Name	Description
pass_completions	Number of completed passes by the quarterback
pass_attempts	Number of pass attempts thrown
passing_yards	Total yards gained through passing
passing_touchdowns	Touchdowns scored via passing plays
interceptions_thrown	Number of passes intercepted by the defense
passing_first_downs	Number of first downs gained by passing
times_sacked	Number of times the quarterback was sacked
sack_yards_lost	Total yards lost due to sacks
pass_fumbles	Number of fumbles on passing plays
pass_fumbles_lost	Fumbles on passing plays that were lost to defense
completion_percentage	Percentage of pass attempts completed
yards_per_attempt	Average yards gained per attempted pass
rush_attempts	Number of rushing attempts
rushing_yards	Total yards gained by rushing
rushing_touchdowns	Touchdowns scored through rushing plays
rushing_first_downs	First downs gained by rushing
rush_fumbles	Number of fumbles on rushing plays
rush_fumbles_lost	Fumbles on rushing plays that were lost
longest_rush	Longest rushing play recorded
yards_per_rush	Average yards gained per rush attempt
receptions	Number of passes caught
targets	Number of pass attempts thrown toward the receiver
receiving_yards	Total yards gained through receiving
receiving_touchdowns	Touchdowns scored on receptions
receiving_first_downs	First downs gained via receptions
receiving_fumbles	Fumbles on completed receptions
receiving_fumbles_lost	Receiving fumbles that were lost

yards_per_reception	Average yards gained per catch
fg_made_(0-19)	Field goals made from 0–19 yards
fg_attempted_(0-19)	Field goal attempts from 0–19 yards
fg_made_(20-29)	Field goals made from 20–29 yards
fg_attempted_(20-29)	Field goal attempts from 20–29 yards
fg_made_(30-39)	Field goals made from 30–39 yards
fg_attempted_(30-39)	Field goal attempts from 30–39 yards
fg_made_(40-49)	Field goals made from 40–49 yards
fg_attempted_(40-49)	Field goal attempts from 40–49 yards
fg_made_(50+)	Field goals made from 50 yards or more
fg_attempted_(50+)	Field goal attempts from 50 yards or more
fg_made_(total)	Total field goals made
fg_attempted_(total)	Total field goal attempts
extra_points_made	Extra points made after touchdowns
extra_points_attempted	Extra point attempts after touchdowns
kickoffs	Number of kickoffs made
kickoff_yards	Total yards on all kickoffs
kickoff_fair_catches	Fair catches made on kickoffs
touchbacks	Number of touchbacks on kickoffs
kick_returns_allowed	Number of kickoff returns allowed by coverage team
kick_return_touchdowns_allowed	Touchdowns allowed on kickoff returns
kick_return_yards_allowed_per_return	Average yards allowed per kickoff return
punts	Number of punts
punt_yards	Total yards gained on punts
punts_inside_20	Number of punts that landed inside the opponent’s 20-yard line

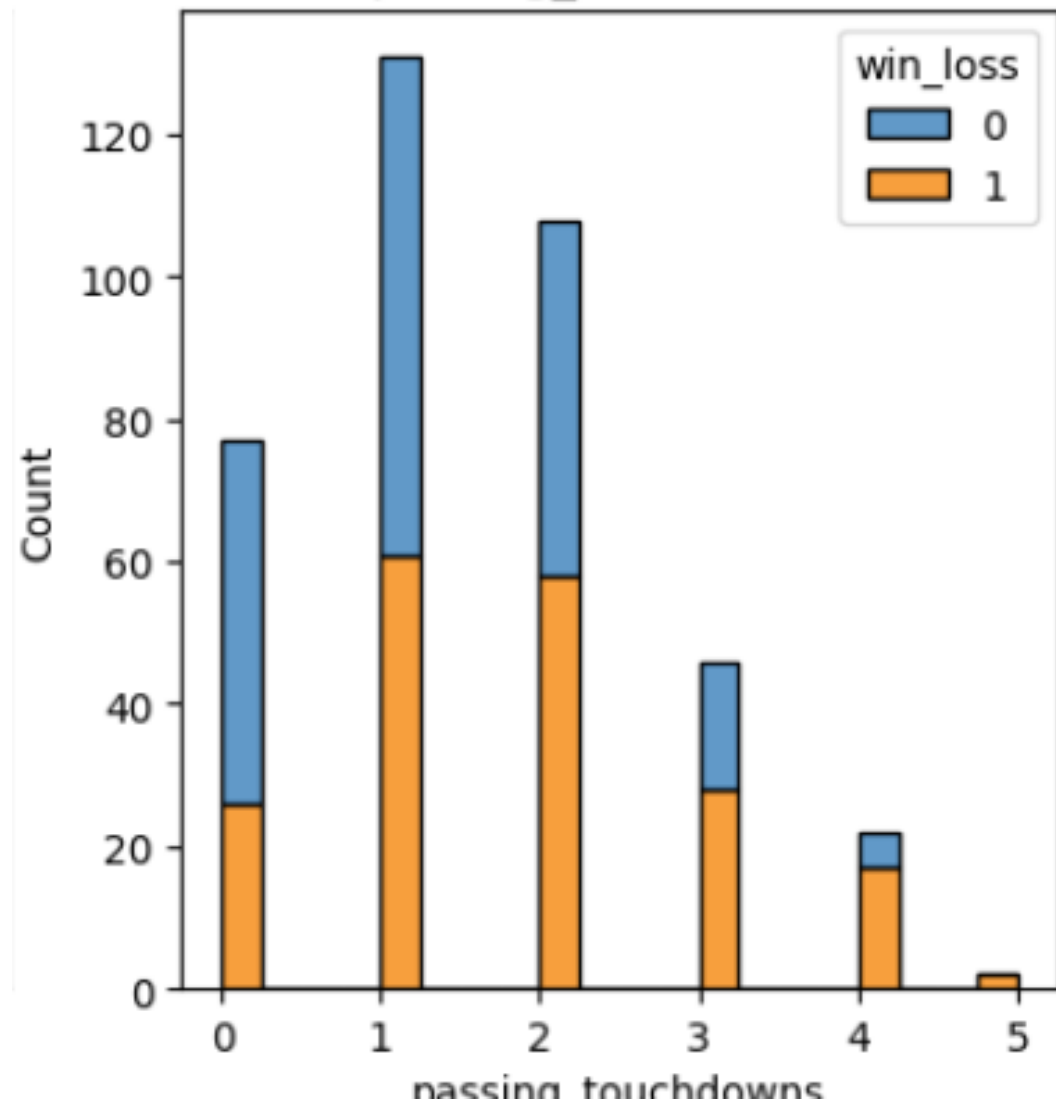
punt_fair_catches	Number of fair catches on punts
punt_touchbacks	Touchbacks resulting from punts
blocked_punts	Number of punts blocked by defense
punt_returns_allowed	Punt returns allowed by coverage team
punt_return_yards_allowed_per_return	Average yards allowed per punt return
kick_returns	Number of kick returns by the team
kick_return_yards	Total yards gained on kick returns
kick_return_touchdowns	Touchdowns scored on kick returns
punt_returns	Number of punt returns
average_punt_return_yards	Average yards gained per punt return
solo_tackles	Number of solo tackles
assisted_tackles	Number of assisted tackles
total_tackles	Total tackles (solo + assisted)
sacks	Number of quarterback sacks
sack_yards	Yards lost by opponent due to sacks
run_stuffs	Number of run plays stopped at or behind line of scrimmage
interceptions	Passes intercepted by defense
interception_return_yards	Yards gained on interception returns
interception_touchdowns	Touchdowns scored on interception returns
forced_fumbles	Fumbles forced on opponents
fumble_return_touchdowns	Touchdowns scored on fumble recoveries
passes_defended	Pass breakups or deflections
safeties	Safeties scored by the defense
team_name_full	Full team name (e.g., “Arizona Cardinals”)

<u>team_abbr</u>	Team abbreviation (e.g., “ARI”)
<u>win_loss</u>	Indicates whether the team won or lost the game (0=loss, 1=win)
<u>game_week</u>	Week number of the NFL season

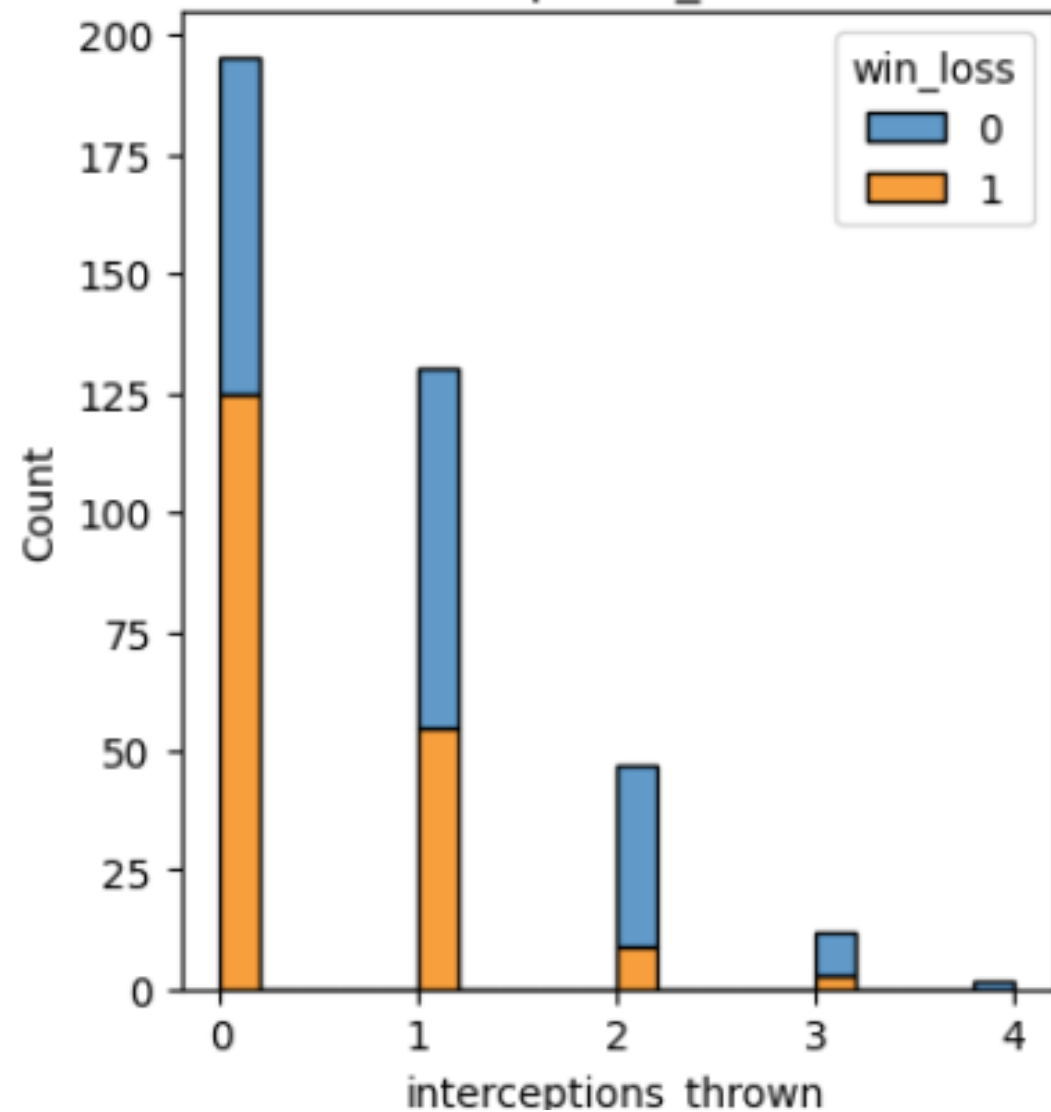
Exploratory Data Analysis

- I created a series of visualizations to explore the distributions of team statistics.
- The goal of this analysis is to identify key statistics that could be the most important factors in predicting wins and losses in the NFL.

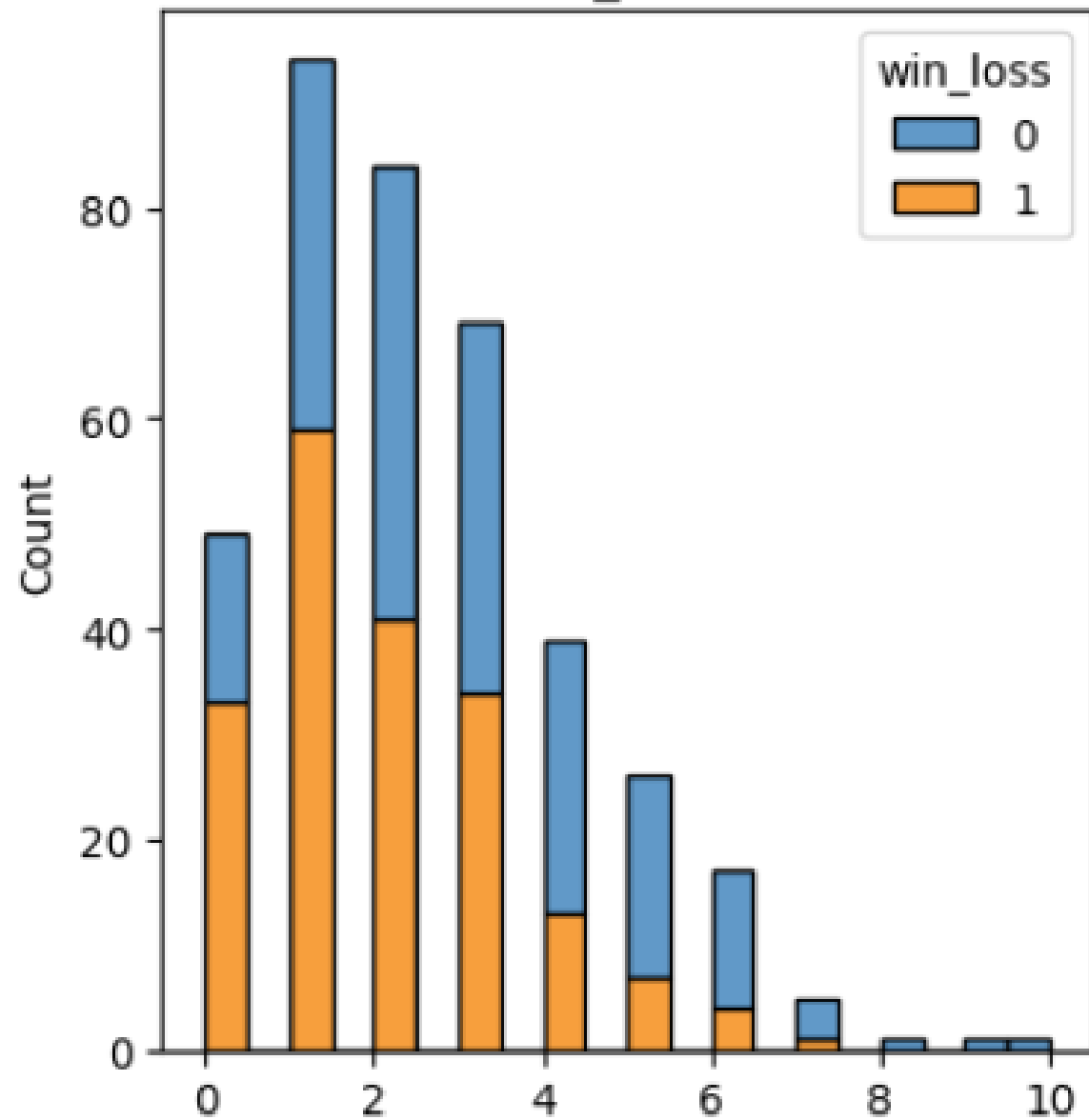
passing_touchdowns



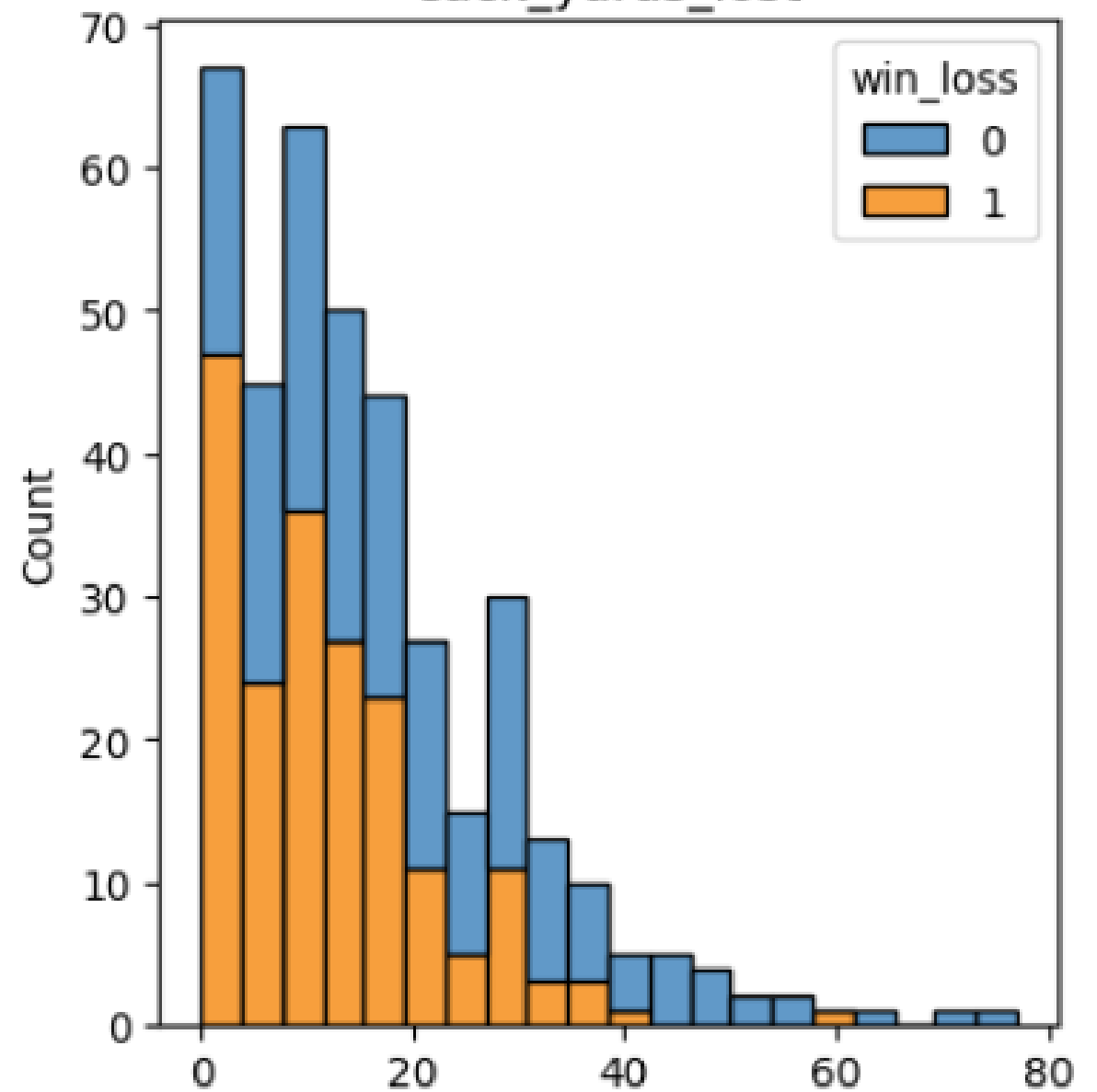
interceptions_thrown



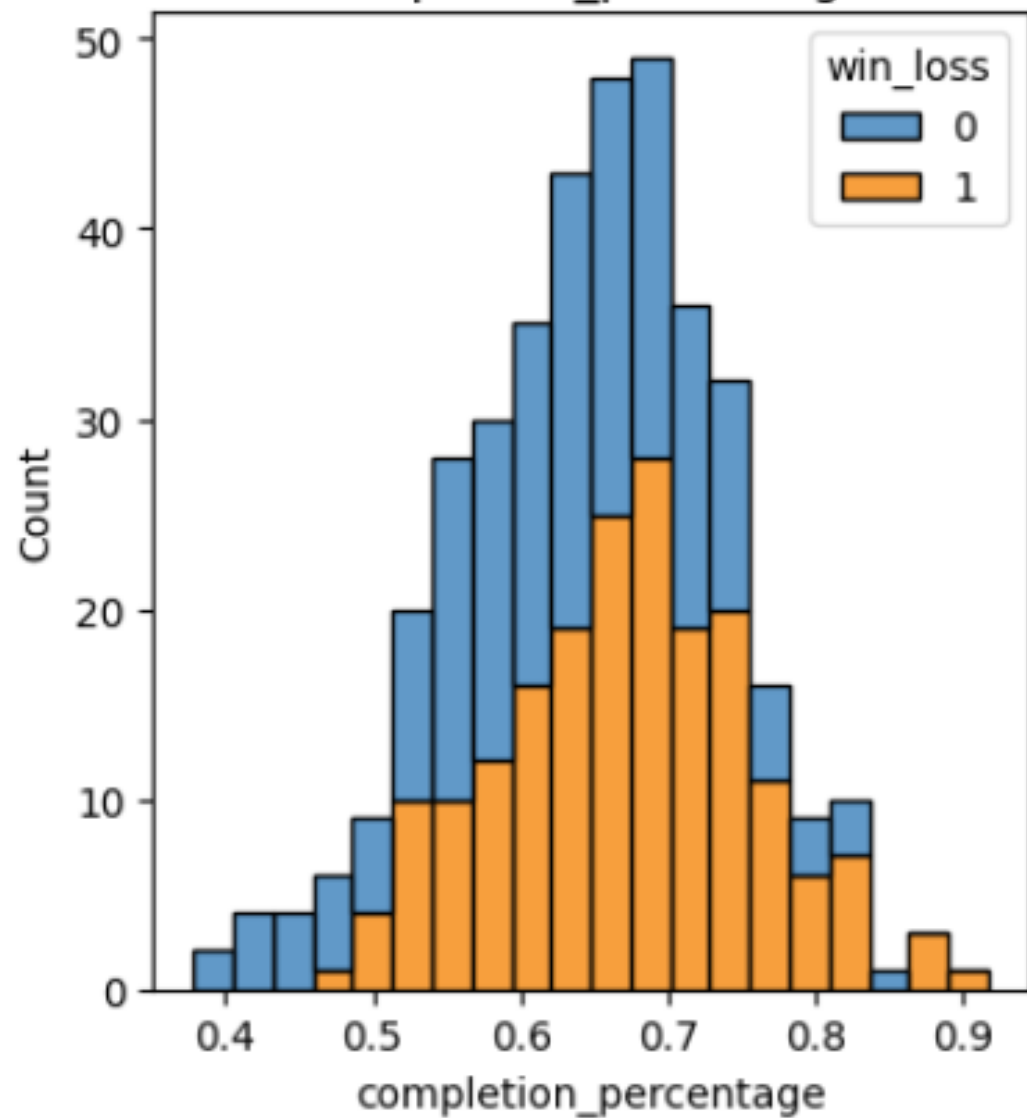
times_sacked



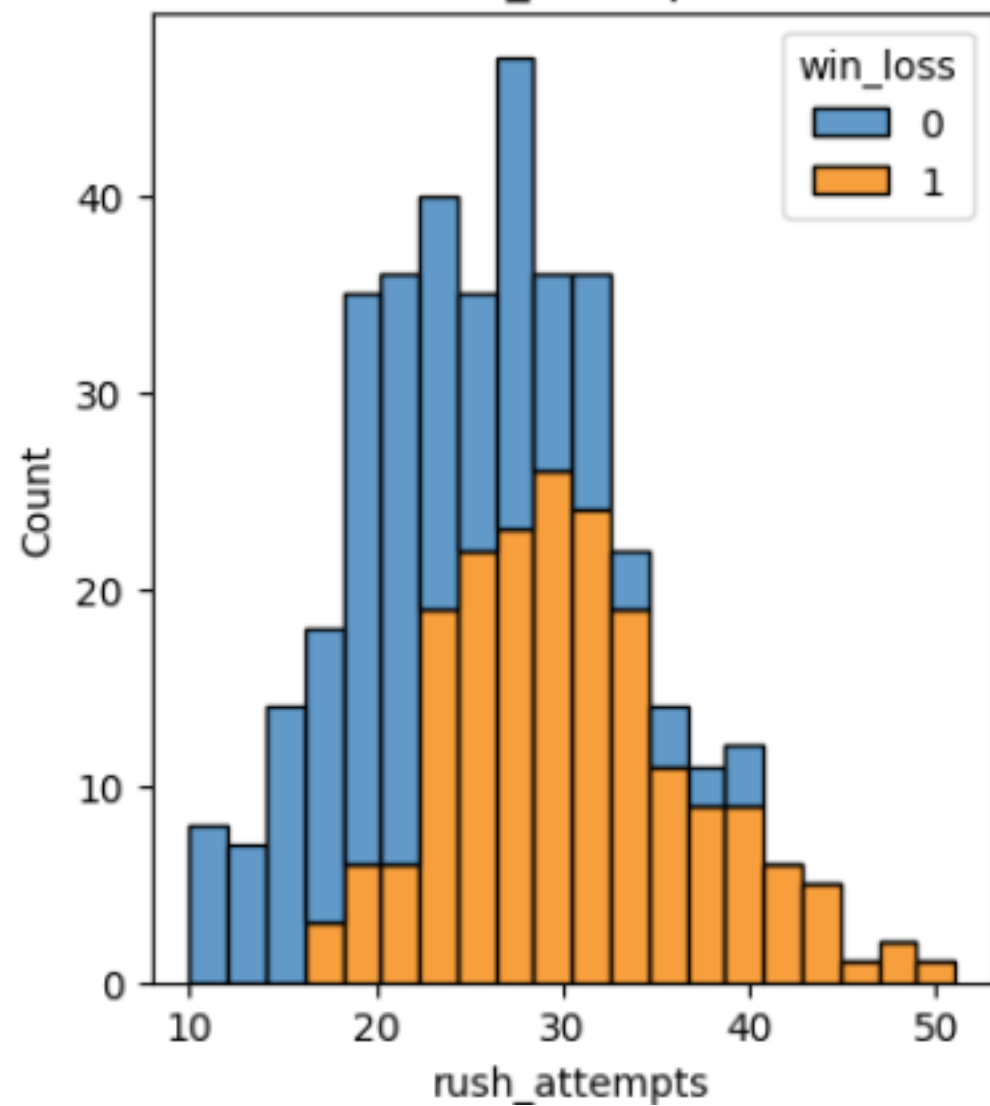
sack_yards_lost



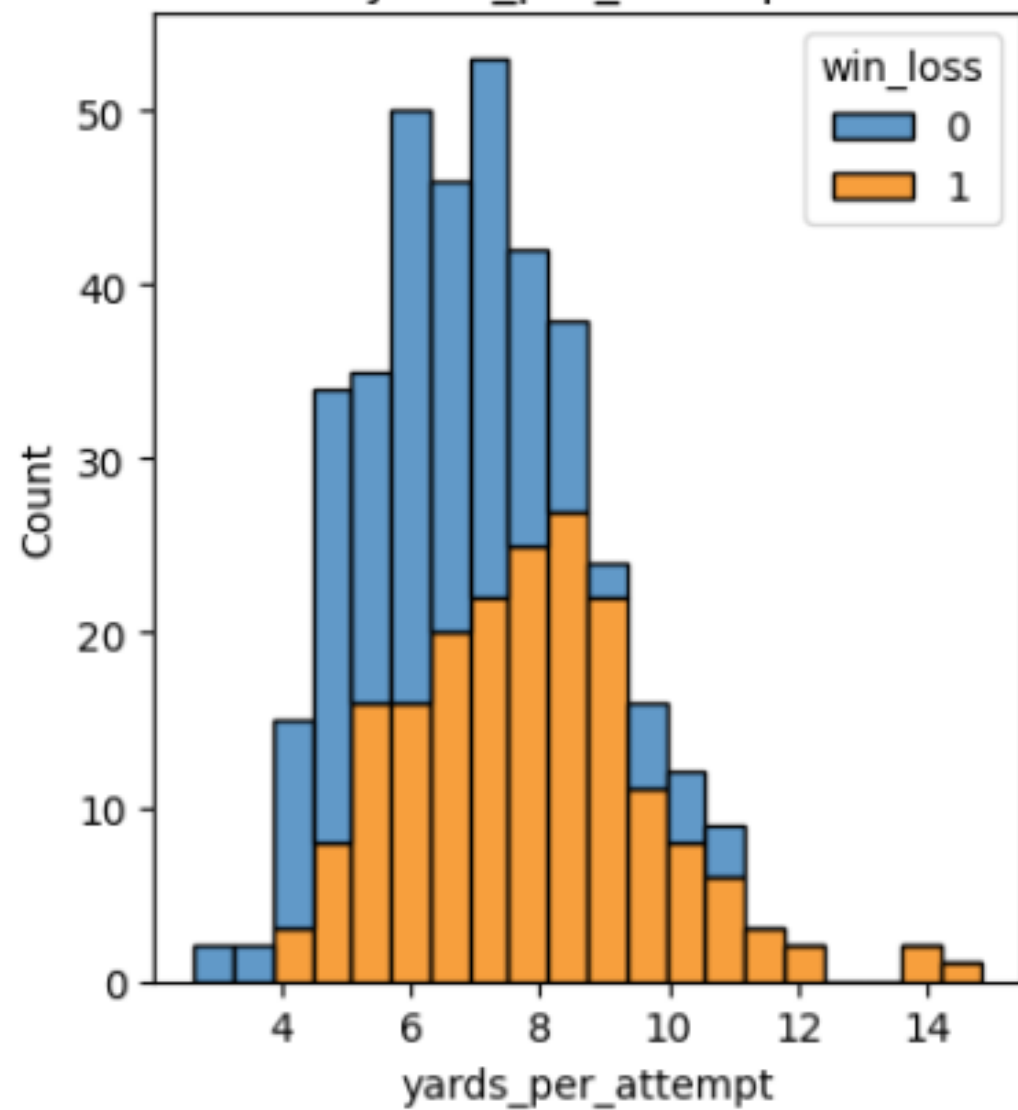
completion_percentage



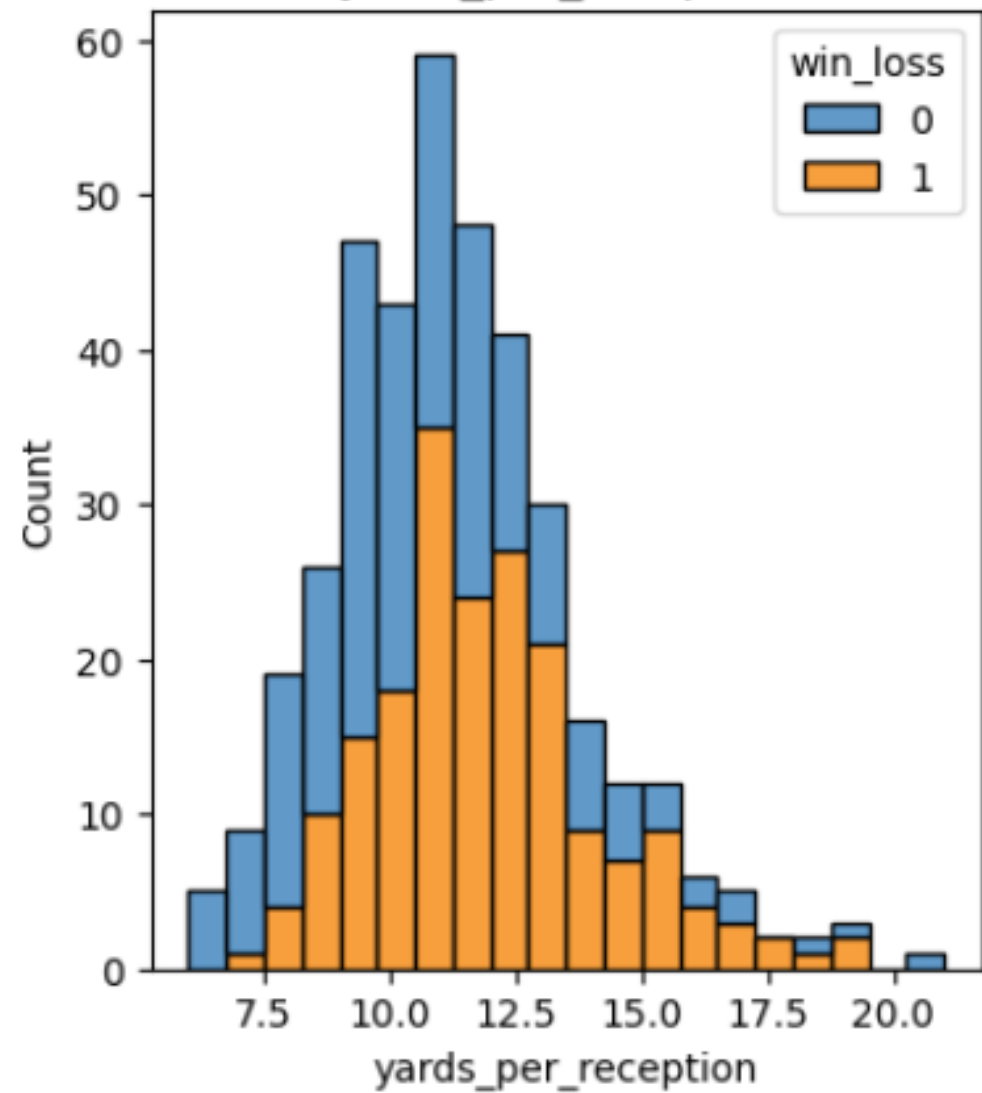
rush_attempts

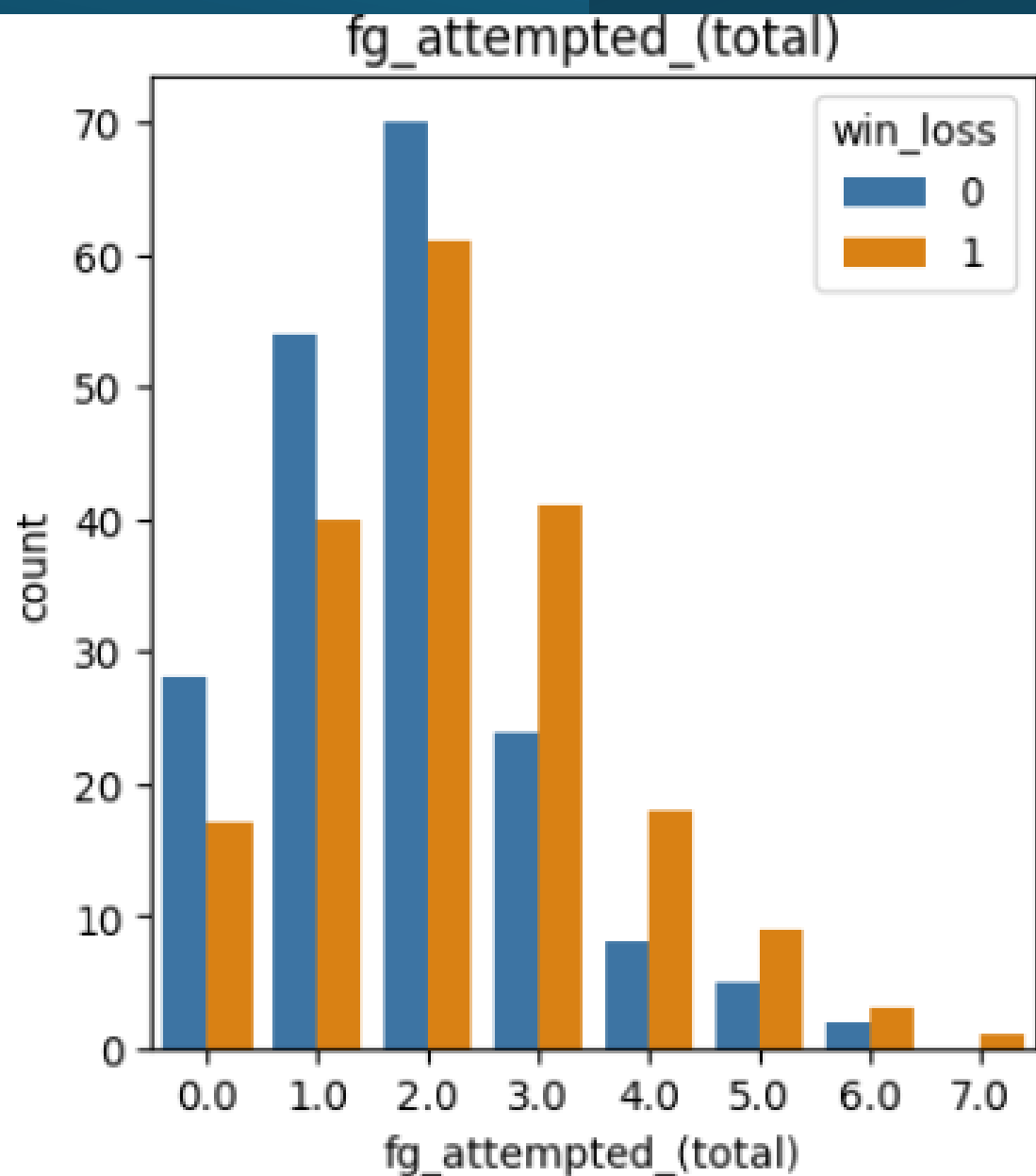
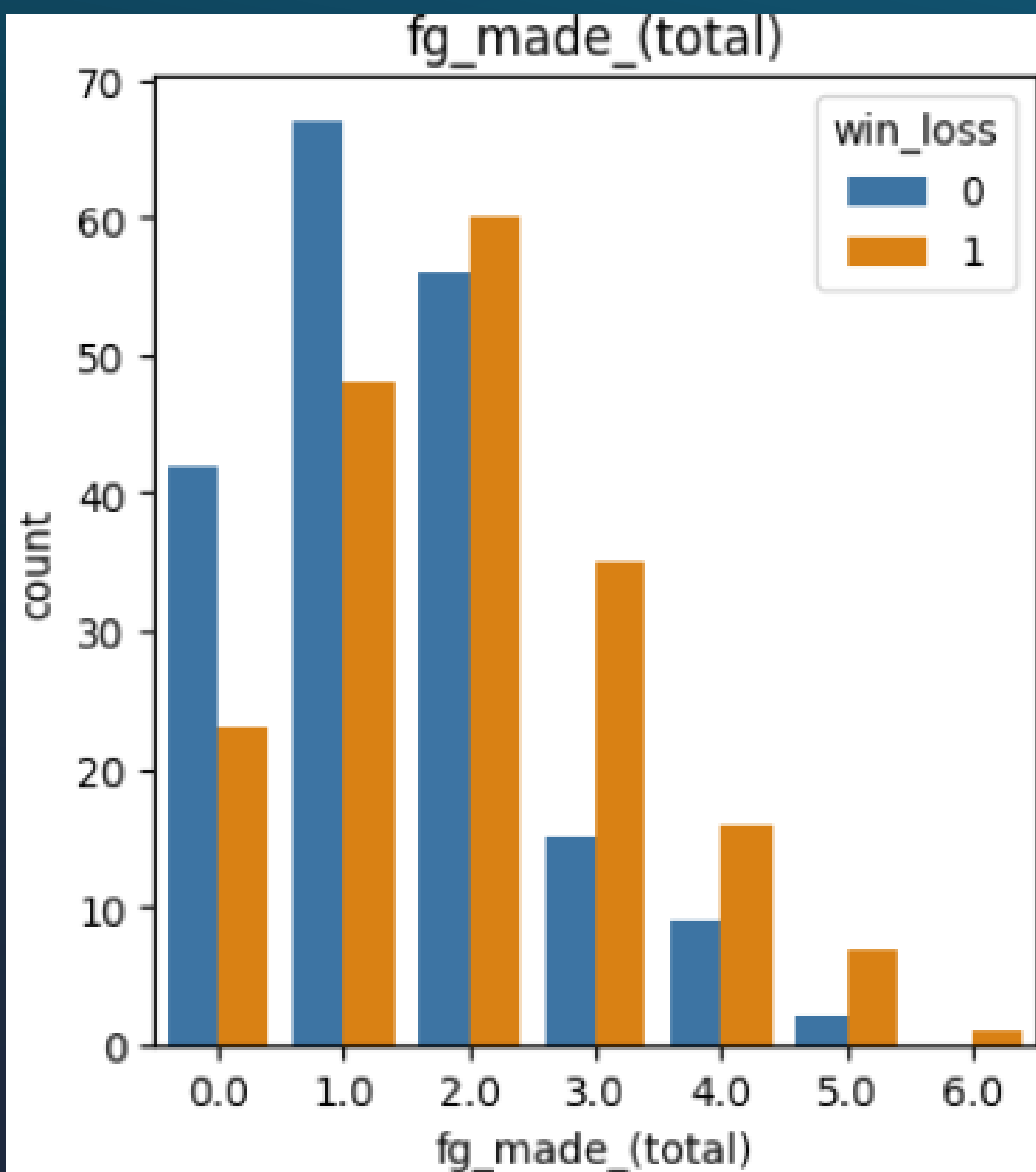


yards_per_attempt

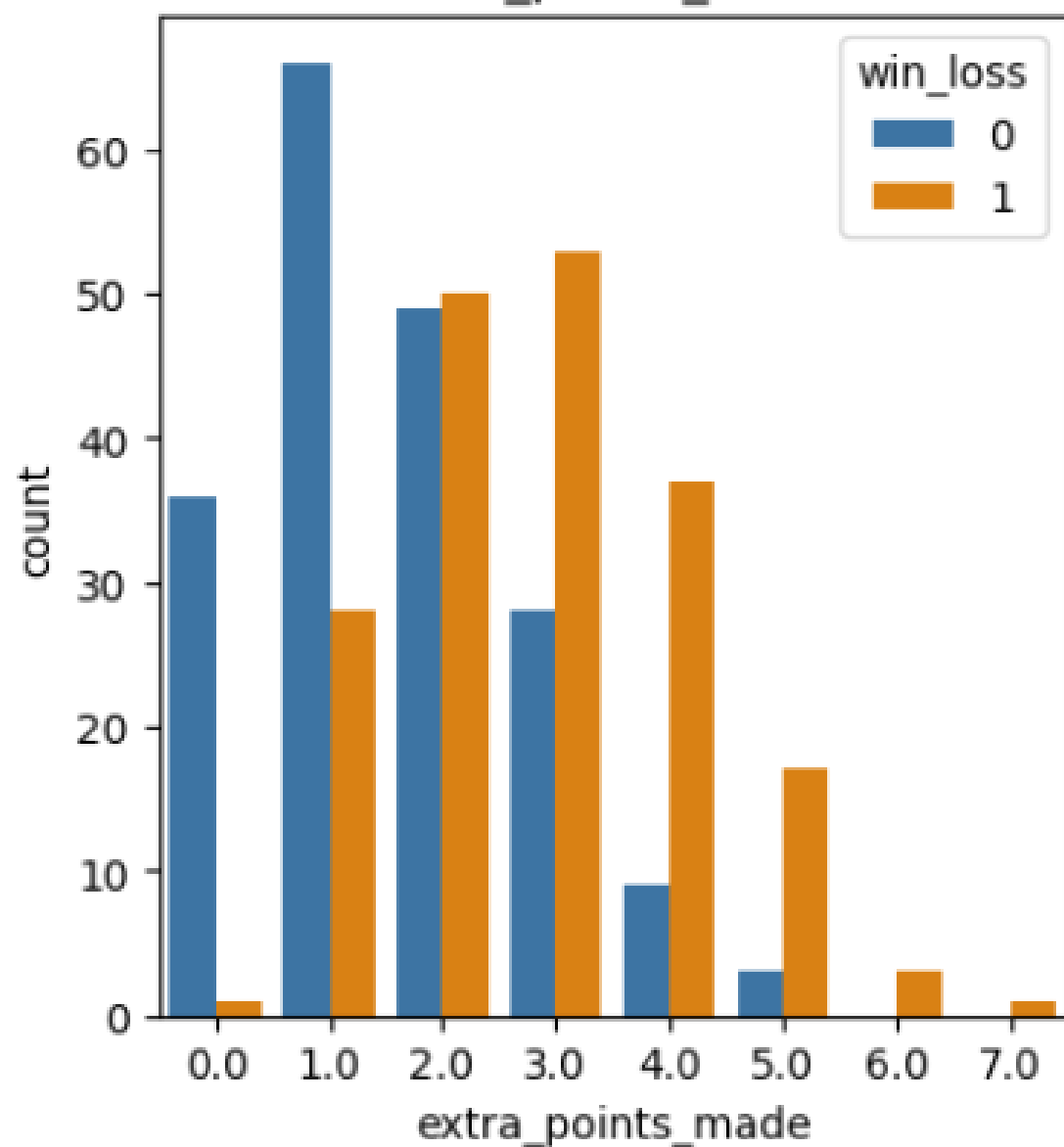


yards_per_reception

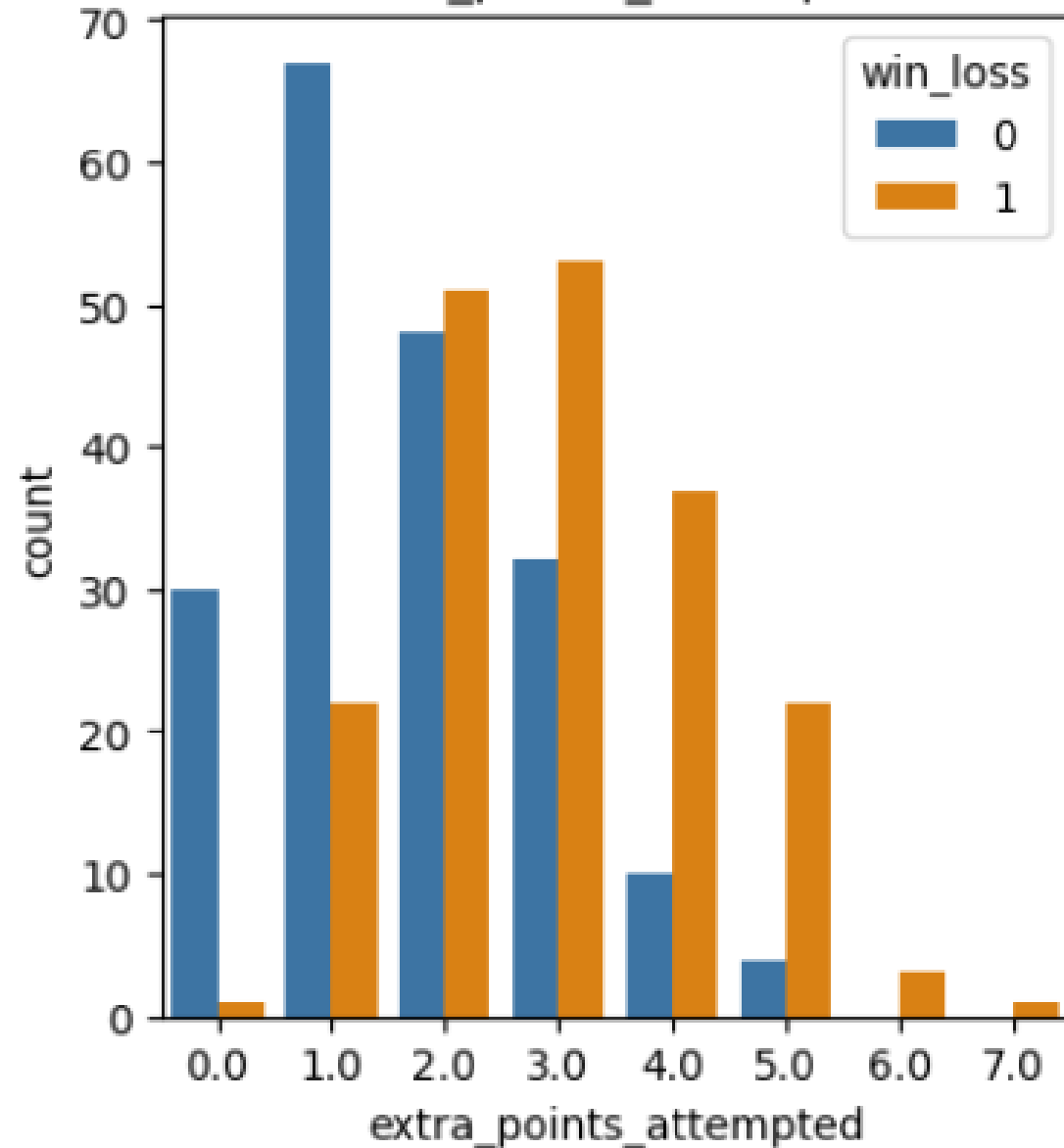




extra_points_made



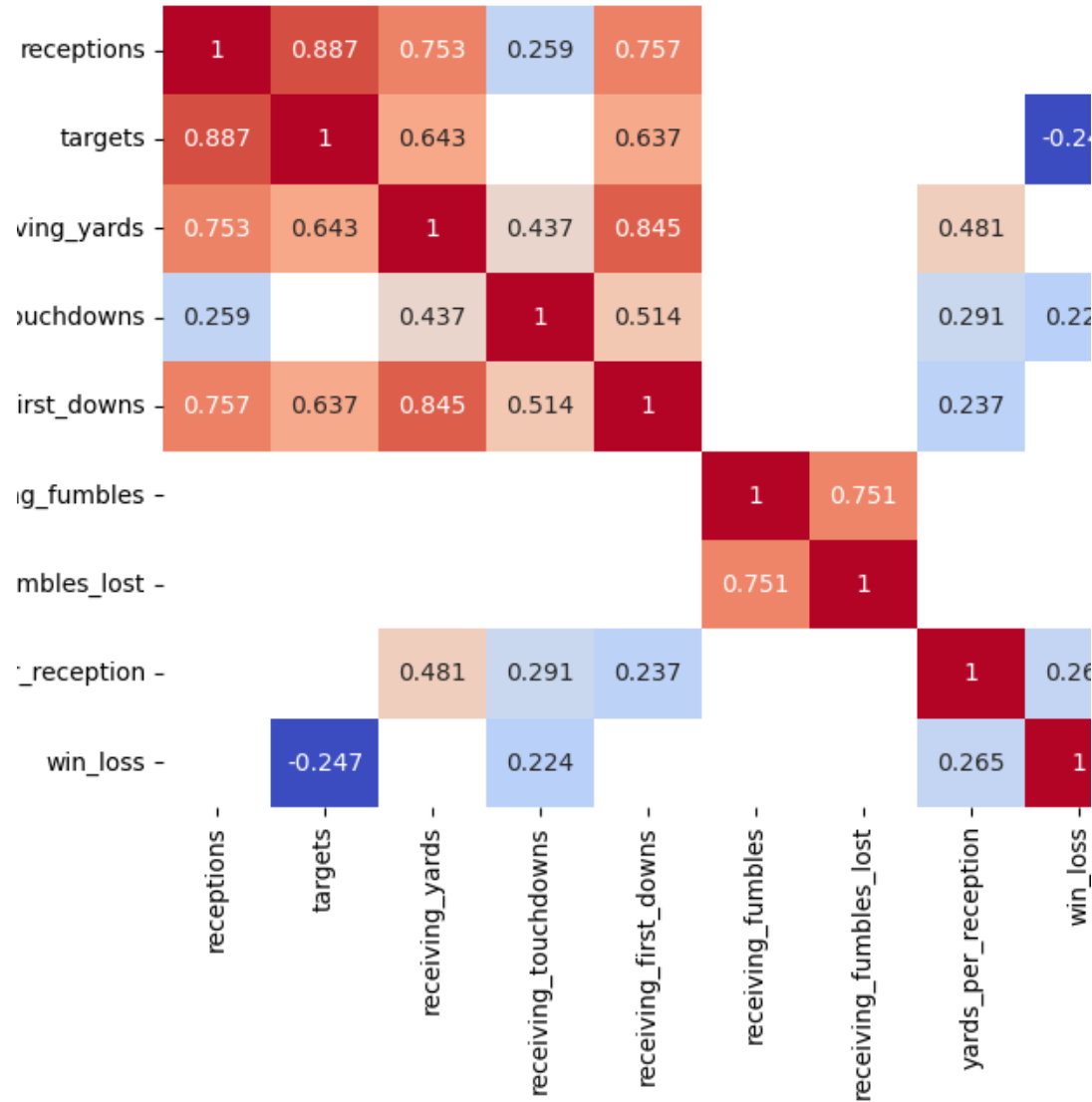
extra_points_attempted



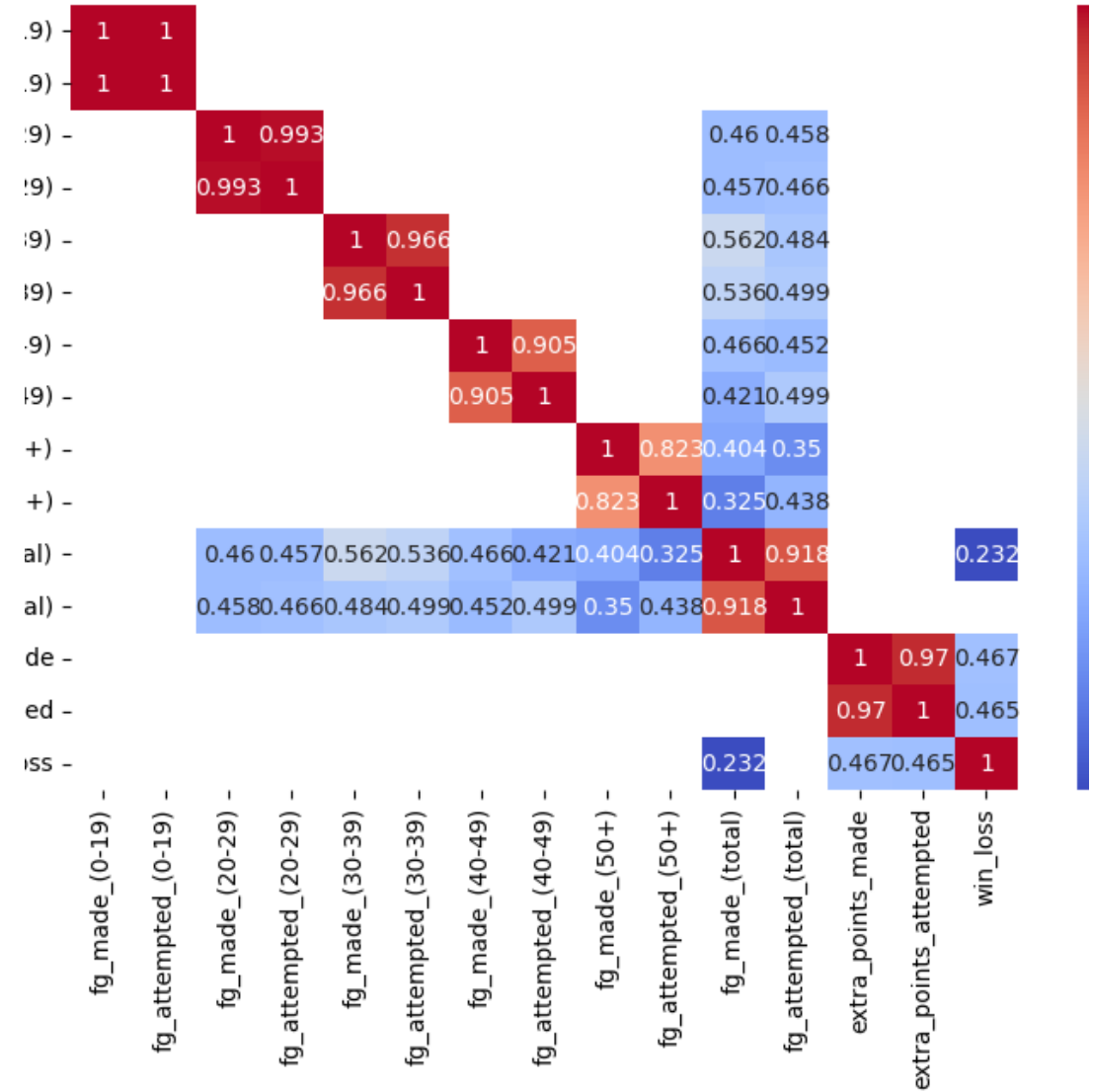
pass_completions	1	0.856	0.738	0.254		0.748				0.371			
pass_attempts	0.856	1	0.603			0.614					-0.25	-0.29	
passing_yards	0.738	0.603	1	0.434		0.843				0.341	0.59		
passing_touchdowns	0.254		0.434	1		0.507				0.352	0.436	0.223	
interceptions_thrown					1							-0.317	
passing_first_downs	0.748	0.614	0.843	0.507		1				0.341	0.388		
times_sacked							1	0.9	0.269	0.239		-0.278	
sack_yards_lost							0.9	1	0.289	0.249		-0.296	
pass_fumbles							0.269	0.289	1	0.629			
pass_fumbles_lost							0.239	0.249	0.629	1		-0.204	
completion_percentage	0.371		0.341	0.352		0.341				1	0.577	0.232	
yards_per_attempt		-0.25	0.59	0.436		0.388				0.577	1	0.36	
win_loss		-0.29		0.223	-0.317		-0.278	-0.296		-0.204	0.232	0.36	1

ts	1	0.701	0.418	0.753		0.216		0.52	
ds	0.701	1	0.477	0.801		0.666	0.731	0.337	
ns	0.418	0.477	1	0.541		0.341	0.275	0.347	
ns	0.753	0.801	0.541	1		0.313	0.428	0.33	
es -					1	0.666			
st -					0.666	1		-0.256	
sh	0.216	0.666	0.341	0.313		1	0.725		
sh -		0.731	0.275	0.428		0.725	1		
ss	0.52	0.337	0.347	0.33		-0.256		1	
	rush_attempts	rushing_yards	rushing_touchdowns	rushing_first_downs	rush_fumbles	rush_fumbles_lost	longest_rush	yards_per_rush	win_loss

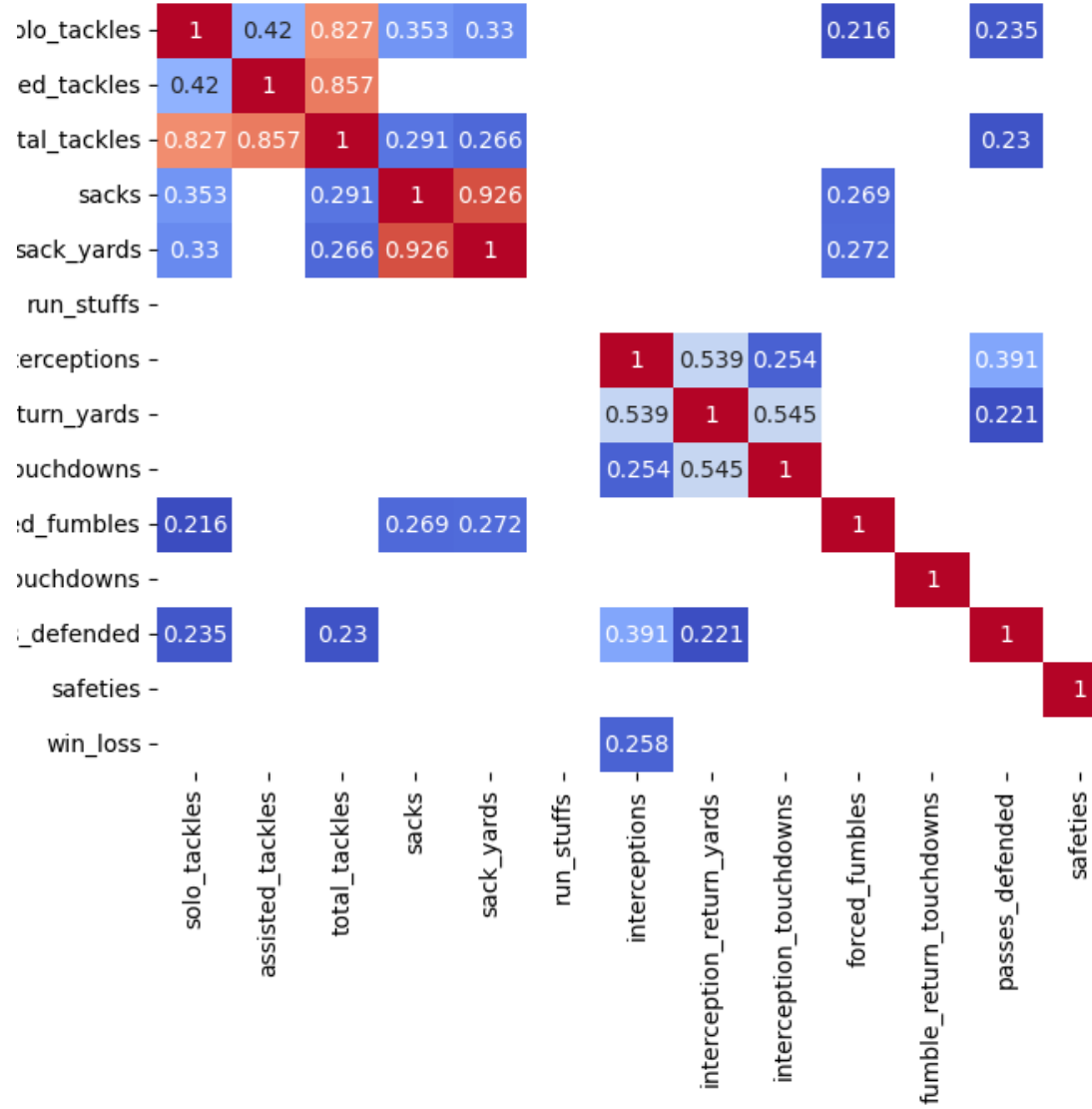
Correlation Matrix for recieving stats



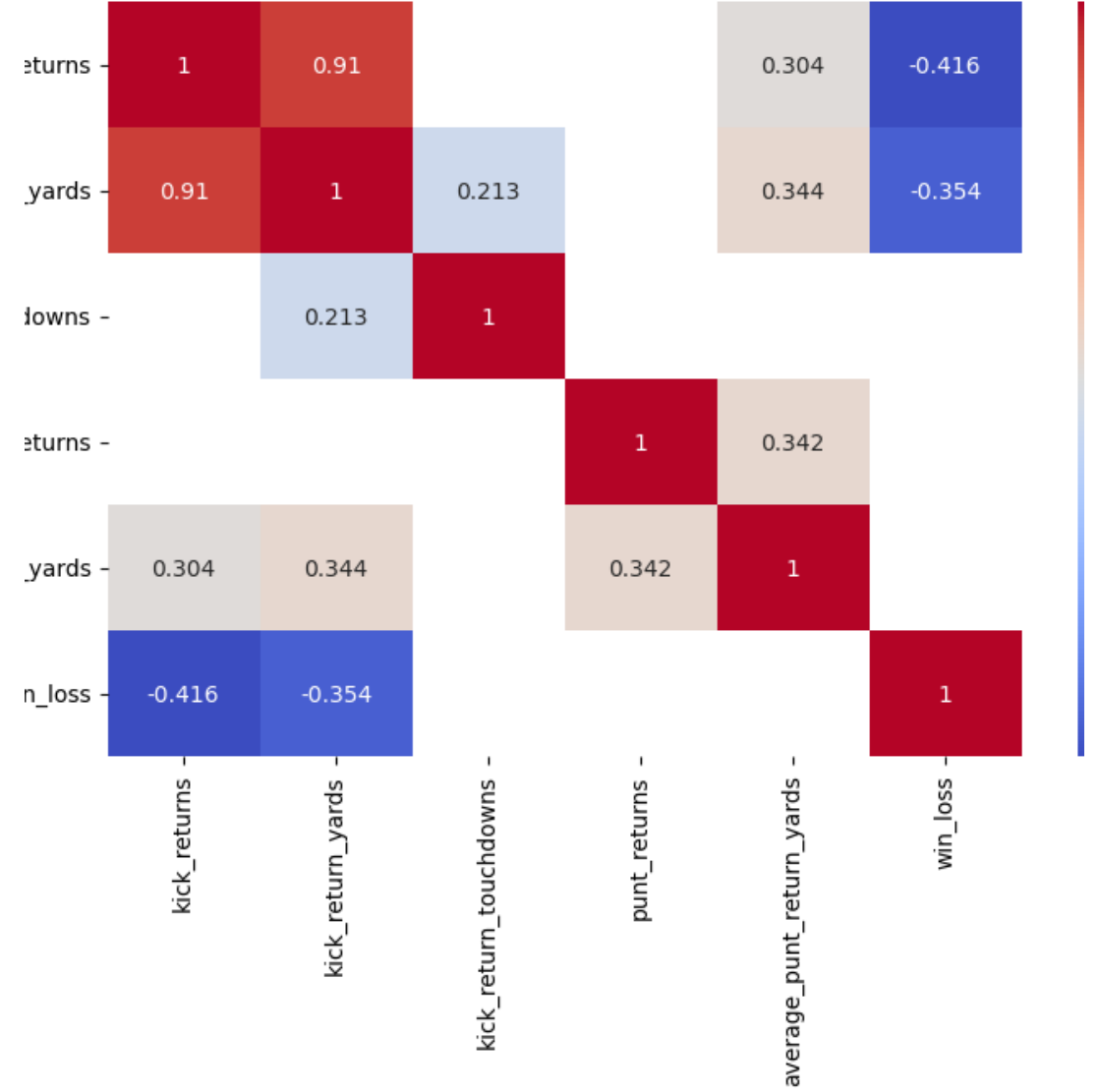
Correlation Matrix for kicking stats



Correlation Matrix for defense stats



Correlation Matrix for return stats



Modeling

- There were missing values in this dataset.
 - Filled with 0.
- Built a logistic regression model on each of the statistics groups.
- After running each model, I would add significant columns to a list to use in my final model.

Logit Regression Results							
Dep. Variable:	win_loss	No. Observations:		386			
Model:	Logit	Df Residuals:		377			
Method:	MLE	Df Model:		8			
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:		0.1746			
Time:	17:35:41	Log-Likelihood:		-220.82			
converged:	True	LL-Null:		-267.55			
Covariance Type:	nonrobust	LLR p-value:		9.240e-17			
	coef	std err	z	P> z	[0.025	0.975]	
const	3.1901	2.065	1.545	0.122	-0.858	7.238	
receptions	-0.1085	0.119	-0.908	0.364	-0.343	0.126	
targets	-0.1580	0.036	-4.383	0.000	-0.229	-0.087	
receiving_yards	0.0118	0.010	1.167	0.243	-0.008	0.032	
receiving_touchdowns	0.1344	0.125	1.078	0.281	-0.110	0.379	
receiving_first_downs	0.2379	0.074	3.218	0.001	0.093	0.383	
receiving_fumbles	-0.2795	0.269	-1.038	0.299	-0.807	0.248	
receiving_fumbles_lost	-0.4092	0.411	-0.997	0.319	-1.214	0.395	
yards_per_reception	-0.1484	0.180	-0.822	0.411	-0.502	0.205	

Logit Regression Results							
Dep. Variable:	win_loss	No. Observations:		386			
Model:	Logit	Df Residuals:		377			
Method:	MLE	Df Model:		8			
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:		0.3222			
Time:	17:34:57	Log-Likelihood:		-181.36			
converged:	True	LL-Null:		-267.55			
Covariance Type:	nonrobust	LLR p-value:		4.077e-33			
	coef	std err	z	P> z	[0.025	0.975]	
const	-5.8326	1.842	-3.167	0.002	-9.442	-2.223	
rush_attempts	0.2767	0.075	3.688	0.000	0.130	0.424	
rushing_yards	-0.0115	0.016	-0.727	0.468	-0.043	0.020	
rushing_touchdowns	0.6327	0.175	3.624	0.000	0.291	0.975	
rushing_first_downs	-0.1782	0.093	-1.916	0.055	-0.360	0.004	
rush_fumbles	-0.0680	0.204	-0.333	0.739	-0.468	0.332	
rush_fumbles_lost	-1.0842	0.322	-3.365	0.001	-1.716	-0.453	
longest_rush	0.0288	0.015	1.867	0.062	-0.001	0.059	
yards_per_rush	0.0454	0.394	0.115	0.908	-0.727	0.817	

Logit Regression Results							
Dep. Variable:	win_loss	No. Observations:		386			
Model:	Logit	Df Residuals:		373			
Method:	MLE	Df Model:		12			
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:		0.2826			
Time:	17:34:50	Log-Likelihood:		-191.94			
converged:	True	LL-Null:		-267.55			
Covariance Type:	nonrobust	LLR p-value:		3.221e-26			
	coef	std err	z	P> z	[0.025	0.975]	
const	-0.9969	4.219	-0.236	0.813	-9.266	7.273	
pass_completions	-0.1830	0.246	-0.743	0.457	-0.666	0.300	
pass_attempts	-0.0099	0.131	-0.075	0.940	-0.266	0.247	
passing_yards	-0.0045	0.012	-0.372	0.710	-0.028	0.019	
passing_touchdowns	-0.0305	0.139	-0.219	0.827	-0.303	0.242	
interceptions_thrown	-0.7151	0.169	-4.228	0.000	-1.047	-0.384	
passing_first_downs	0.2911	0.081	3.597	0.000	0.132	0.450	
times_sacked	-0.0848	0.166	-0.512	0.609	-0.409	0.240	
sack_yards_lost	-0.0326	0.024	-1.370	0.171	-0.079	0.014	
pass_fumbles	0.3221	0.242	1.330	0.183	-0.153	0.797	
pass_fumbles_lost	-1.2281	0.406	-3.027	0.002	-2.023	-0.433	
completion_percentage	3.8204	7.913	0.483	0.629	-11.689	19.330	
yards_per_attempt	0.2538	0.361	0.704	0.482	-0.453	0.961	

Logit Regression Results							
Dep. Variable:	win_loss	No. Observations:		386			
Model:	Logit	Df Residuals:		374			
Method:	MLE	Df Model:		11			
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:		0.2699			
Time:	17:35:47	Log-Likelihood:		-195.33			
converged:	True	LL-Null:		-267.55			
Covariance Type:	nonrobust	LLR p-value:		2.036e-25			
	coef	std err	z	P> z	[0.025	0.975]	
const	-3.3497	0.412	-8.126	0.000	-4.158	-2.542	
fg_made_(0-19)	0.0003	nan	nan	nan	nan	nan	
fg_attempted_(0-19)	0.0003	nan	nan	nan	nan	nan	
fg_made_(20-29)	0.6892	0	inf	0.000	0.689	0.689	
fg_attempted_(20-29)	-0.2803	1.08e+07	-2.59e-08	1.000	-2.12e+07	2.12e+07	
fg_made_(30-39)	0.5073	1.873	0.271	0.787	-3.164	4.178	
fg_attempted_(30-39)	-0.2813	1.08e+07	-2.6e-08	1.000	-2.12e+07	2.12e+07	
fg_made_(40-49)	-0.3309	1.529	-0.216	0.829	-3.327	2.665	
fg_attempted_(40-49)	0.1389	1.08e+07	1.28e-08	1.000	-2.12e+07	2.12e+07	
fg_made_(50+)	0.0658	1.583	0.042	0.967	-3.037	3.169	
fg_attempted_(50+)	0.0366	1.08e+07	3.38e-09	1.000	-2.12e+07	2.12e+07	
fg_made_(total)	0.9316	1.626	0.573	0.567	-2.256	4.119	
fg_attempted_(total)	-0.3858	1.08e+07	-3.57e-08	1.000	-2.12e+07	2.12e+07	
extra_points_made	0.2957	0.369	0.802	0.422	-0.427	1.018	
extra_points_attempted	0.7258	0.372	1.951	0.051	-0.003	1.455	

Logit Regression Results							
Dep. Variable:	win_loss	No. Observations:	386				
Model:	Logit	Df Residuals:	382				
Method:	MLE	Df Model:	3				
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:	0.2323				
Time:	17:35:56	Log-Likelihood:	-205.40				
converged:	True	LL-Null:	-267.55				
Covariance Type:	nonrobust	LLR p-value:	9.165e-27				
	coef	std err	z	P> z	[0.025	0.975]	
const	-3.8232	0.481	-7.951	0.000	-4.766	-2.881	
kickoffs	-0.5393	0.346	-1.560	0.119	-1.217	0.138	
kickoff_yards	0.0214	0.006	3.722	0.000	0.010	0.033	
touchbacks	0.0653	0.129	0.506	0.613	-0.187	0.318	

Logit Regression Results							
Dep. Variable:	win_loss	No. Observations:	386				
Model:	Logit	Df Residuals:	380				
Method:	MLE	Df Model:	5				
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:	0.1374				
Time:	17:36:12	Log-Likelihood:	-230.78				
converged:	True	LL-Null:	-267.55				
Covariance Type:	nonrobust	LLR p-value:	1.878e-14				
	coef	std err	z	P> z	[0.025	0.975]	
const	2.3508	0.338	6.961	0.000	1.689	3.013	
kick_returns	-0.7337	0.181	-4.062	0.000	-1.088	-0.380	
kick_return_yards	0.0054	0.006	0.831	0.406	-0.007	0.018	
kick_return_touchdowns	1.1431	1.033	1.106	0.269	-0.882	3.168	
punt_returns	-0.1121	0.098	-1.144	0.252	-0.304	0.080	
average_punt_return_yards	0.0020	0.003	0.727	0.467	-0.003	0.007	

Logit Regression Results							
Dep. Variable:	win_loss	No. Observations:	386				
Model:	Logit	Df Residuals:	373				
Method:	MLE	Df Model:	12				
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:	0.1538				
Time:	17:36:19	Log-Likelihood:	-226.41				
converged:	True	LL-Null:	-267.55				
Covariance Type:	nonrobust	LLR p-value:	1.517e-12				
	coef	std err	z	P> z	[0.025	0.975]	
const	1.0072	0.410	2.456	0.014	0.204	1.811	
solo_tackles	-0.0280	nan	nan	nan	nan	nan	
assisted_tackles	-0.0090	nan	nan	nan	nan	nan	
total_tackles	-0.0369	nan	nan	nan	nan	nan	
sacks	-0.2126	0.233	-0.913	0.361	-0.669	0.244	
sack_yards	0.0756	0.032	2.398	0.016	0.014	0.137	
interceptions	0.7278	0.253	2.879	0.004	0.232	1.223	
interception_return_yards	0.0277	0.016	1.704	0.088	-0.004	0.060	
interception_touchdowns	-1.9529	0.967	-2.020	0.043	-3.848	-0.058	
forced_fumbles	0.3940	0.208	1.894	0.058	-0.014	0.802	
fumble_return_touchdowns	1.0549	1.202	0.877	0.380	-1.302	3.412	
passes_defended	0.2012	0.073	2.758	0.006	0.058	0.344	
safeties	0.2648	1.366	0.194	0.846	-2.412	2.942	

Logit Regression Results							
Dep. Variable:	win_loss	No. Observations:	386				
Model:	Logit	Df Residuals:	380				
Method:	MLE	Df Model:	5				
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:	0.01309				
Time:	17:36:04	Log-Likelihood:	-264.05				
converged:	True	LL-Null:	-267.55				
Covariance Type:	nonrobust	LLR p-value:	0.2202				
	coef	std err	z	P> z	[0.025	0.975]	
const	0.4065	0.242	1.680	0.093	-0.068	0.881	
punts	-0.4747	0.289	-1.643	0.100	-1.041	0.092	
punt_yards	0.0062	0.006	1.085	0.278	-0.005	0.017	
punts_inside_20	0.1542	0.112	1.374	0.169	-0.066	0.374	
punt_touchbacks	0.1120	0.218	0.515	0.607	-0.315	0.539	
blocked_punts	-0.4093	0.884	-0.463	0.643	-2.142	1.324	

Feature Selection Summary

interceptions_thrown

passing_first_downs

pass_fumbles_lost

rush_attempts

rushing_touchdowns

rush_fumbles_lost

targets

receiving_first_downs

fg_made_(20-29)

kickoff_yards

kick_returns

sack_yards

interceptions

interception_touchdowns

passes_defended

Logit Regression Results						
Dep. Variable:	win_loss	No. Observations:	386			
Model:	Logit	Df Residuals:	370			
Method:	MLE	Df Model:	15			
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:	0.5364			
Time:	17:43:43	Log-Likelihood:	-124.03			
converged:	True	LL-Null:	-267.55			
Covariance Type:	nonrobust	LLR p-value:	2.750e-52			
	coef	std err	z	P> z	[0.025	0.975]
const	-4.0886	1.434	-2.851	0.004	-6.899	-1.278
interceptions_thrown	-0.5415	0.224	-2.421	0.015	-0.980	-0.103
passing_first_downs	-0.5092	0.545	-0.935	0.350	-1.577	0.558
pass_fumbles_lost	-1.0107	0.620	-1.630	0.103	-2.226	0.205
rush_attempts	0.1375	0.032	4.292	0.000	0.075	0.200
rushing_touchdowns	0.2939	0.193	1.526	0.127	-0.083	0.671
rush_fumbles_lost	-0.2232	0.431	-0.518	0.605	-1.068	0.622
targets	-0.1101	0.036	-3.049	0.002	-0.181	-0.039
receiving_first_downs	0.8015	0.549	1.460	0.144	-0.275	1.878
fg_made_(20-29)	-0.0679	0.278	-0.245	0.807	-0.612	0.476
kickoff_yards	0.0075	0.002	3.545	0.000	0.003	0.012
kick_returns	-0.6198	0.115	-5.376	0.000	-0.846	-0.394
sack_yards	0.0479	0.018	2.713	0.007	0.013	0.082
interceptions	0.9244	0.316	2.924	0.003	0.305	1.544
interception_touchdowns	-1.3758	1.251	-1.099	0.272	-3.828	1.077
passes_defended	0.0952	0.094	1.011	0.312	-0.089	0.280

Logit Regression Results						
Dep. Variable:	win_loss	No. Observations:	386			
Model:	Logit	Df Residuals:	376			
Method:	MLE	Df Model:	9			
Date:	Mon, 08 Dec 2025	Pseudo R-squ.:	0.5232			
Time:	17:52:17	Log-Likelihood:	-127.57			
converged:	True	LL-Null:	-267.55			
Covariance Type:	nonrobust	LLR p-value:	4.616e-55			
	coef	std err	z	P> z	[0.025	0.975]
const	-4.7289	1.298	-3.644	0.000	-7.273	-2.185
interceptions_thrown	-0.5463	0.220	-2.487	0.013	-0.977	-0.116
pass_fumbles_lost	-1.1304	0.426	-2.653	0.008	-1.965	-0.295
rush_attempts	0.1564	0.029	5.325	0.000	0.099	0.214
targets	-0.1040	0.034	-3.062	0.002	-0.171	-0.037
receiving_first_downs	0.2868	0.071	4.025	0.000	0.147	0.426
kickoff_yards	0.0084	0.002	4.322	0.000	0.005	0.012
kick_returns	-0.5976	0.112	-5.348	0.000	-0.817	-0.379
sack_yards	0.0497	0.017	2.890	0.004	0.016	0.083
interceptions	0.9724	0.284	3.419	0.001	0.415	1.530

Testing Other Models

Logistic Regression Accuracy: 0.8846153846153846
False Positive Rate: 15.384615384615385
False Negative Rate: 7.6923076923076925

Random Forest Accuracy: 0.8589743589743589
False Positive Rate: 17.94871794871795
False Negative Rate: 10.256410256410255

XGBoost Accuracy: 0.8205128205128205
False Positive Rate: 17.94871794871795
False Negative Rate: 17.94871794871795

Naive Bayes Accuracy: 0.8846153846153846
False Positive Rate: 17.94871794871795
False Negative Rate: 5.128205128205128

Logistic Regression Accuracy: 0.8461538461538461
False Positive Rate: 17.94871794871795
False Negative Rate: 12.82051282051282

Random Forest Accuracy: 0.8205128205128205
False Positive Rate: 17.94871794871795
False Negative Rate: 17.94871794871795

XGBoost Accuracy: 0.8076923076923077
False Positive Rate: 17.94871794871795
False Negative Rate: 20.51282051282051

Naive Bayes Accuracy: 0.8589743589743589
False Positive Rate: 17.94871794871795
False Negative Rate: 10.256410256410255

Conclusion

- Utilized exploratory data analysis to familiarize and identify certain features that could be an indicator to predict if an NFL team will win or lose.
- Modeled these features and identified significant features using logistic regression.
- Build four models on two different feature sets that resulted in strong accuracy score with decently low false positive and negative rates.

Further Study

- I would collect more data.
 - I wanted to use current data.
 - The downside is having a small dataset.
- I would also test out more models.
- Feature engineering with correlated features.

Data Source

- https://sports.yahoo.com/nfl/stats/weekly/?selectedTable=0&week={%22week%22:5,%22seasonPhase%22:%22REGULAR_SEASON%22}